

1st Half

$$\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R} \text{ s.t. : } \|\vec{x}\| \geq 0, \|\lambda\vec{x}\| = |\lambda|\|\vec{x}\|, \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$$

$$|\vec{x}^T \vec{y}| \leq \|\vec{x}\|_2 \|\vec{y}\|_2 \text{ (C.S.) } \mathbb{R}_+^n = \{\mathbb{R}^n \geq 0\} \quad \mathbb{R}_{++}^n = \{\mathbb{R}^n > 0\}$$

$$\Delta_n = \{\vec{x} \in \mathbb{R}^n : x_i \geq 0 \forall i \in \{1, \dots, n\}, \vec{1}^T \vec{x} = 1\}$$

$$B(\vec{c}, r) = \{\vec{x} \in \mathbb{R}^n : \|\vec{x} - \vec{c}\|_p < r\} \quad // \text{ open } (), \text{ closed } []$$

$$\|\vec{x}\|_p \triangleq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \quad \|A\|_F = \sqrt{\sum_i \sum_j |A_{ij}|^2}$$

$$\|A\vec{x}\|_b \leq \|A\|_{a,b} \|\vec{x}\|_a \quad \text{w/} \quad \|A\|_{a,b} = \max_{\vec{x} \in \mathbb{R}^n} \{\|A\vec{x}\|_b : \|\vec{x}\|_a \leq 1\}$$

$$\|A\|_2 = \|A\|_{2,2} = \sqrt{\lambda_{\max}(A^T A)} = \sigma_{\max}(A) \quad // \text{ spectral norm}$$

$$\|A\|_1 = \max_{i=1, \dots, n} \{\text{col. sums}\} \quad \|A\|_{\infty} = \max_{j=1, \dots, n} \{\text{row. sums}\}$$

$$R_A(\vec{x} \neq 0) = \frac{\vec{x}^T A \vec{x}}{\vec{x}^T \vec{x}} \quad \lambda_{\min}(A) \leq R_A(\vec{x} \neq 0) \leq \lambda_{\max}(A)$$

$$\lambda_{\min}(A) = \min_{\vec{x} \neq 0} R_A(\vec{x}) \quad \lambda_{\max}(A) = \max_{\vec{x} \neq 0} R_A(\vec{x})$$

$$f'(\vec{x}; \vec{d}_i) = \lim_{t \rightarrow 0^+} \frac{f(\vec{x} + t\vec{d}_i) - f(\vec{x})}{t} = \nabla f(\vec{x})^T \vec{d} \quad // \text{ dir. deriv.}$$

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n \quad \nabla^2 f(\vec{x}) = \begin{bmatrix} | & & | \\ \nabla f_1 & \dots & \nabla f_n \\ | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

$$\text{TS } f(\vec{y}) = f(\vec{x}) + \nabla f(\vec{x})^T (\vec{y} - \vec{x}) + \underbrace{\frac{1}{2}(\vec{y} - \vec{x})^T \nabla^2 f(\vec{x}) (\vec{y} - \vec{x})}_{=o\|\vec{y}-\vec{x}\| \text{ for lin. approx}} + \underbrace{o(\|\vec{y} - \vec{x}\|^2)}_{\text{quad. approx}}$$

$$f(\vec{x}^* + \vec{d}) = f(\vec{x}^*) + \underbrace{\nabla f(\vec{x}^*)^T \vec{d}}_{=0 \text{ if stat. pt.}} + \frac{1}{2} \vec{d}^T \nabla^2 f(\vec{x}^*) \vec{d} + o(\|\vec{d}\|^2)$$

$$\lim_{\|\vec{x}\| \rightarrow \infty} f(\vec{x}) = \infty \implies \nabla^2 f(\vec{x}) \succ 0 \quad // \text{ Coercive}$$

Necessary 2nd Order Optimality

$$\text{if } \vec{x}^* \text{ is } \mathbf{local} \text{ min, then } \nabla^2 f(\vec{x}^*) \succeq 0 \quad \mathbf{local} \text{ max if } \preceq 0$$

Sufficient 2nd Order Optimality

$$\text{if } \nabla^2 f(\vec{x}^*) \succ 0 \text{ then } \vec{x}^* \text{ is } \mathbf{strict local} \text{ min} \quad \text{max if } \prec 0$$

$$\text{if } \nabla^2 f(\vec{x}^*) \text{ is indef. then } \vec{x}^* \text{ is a } \mathbf{saddle}$$

Global Optimality

$$\text{if } \nabla^2 f(\vec{x}^*) \succeq 0, \forall \vec{x}, \vec{x}^* = \text{stat. pt.}, \text{ then } \vec{x}^* \text{ is } \mathbf{global} \text{ min}$$

$$\text{if } \exists m > 0 \text{ s.t. } \nabla^2 f(\vec{x}) \succeq mI, \forall \vec{x} \text{ then:}$$

$$f(\vec{x}) \geq f(\vec{x}^*) + \frac{m}{2} \underbrace{\|\vec{x} - \vec{x}^*\|^2}_{\text{bowl above } f(\vec{x}^*)} \geq f(\vec{x}^*)$$

$$\underbrace{A\vec{x} = \vec{b}}_{\text{Least Squares}} \quad A \in \mathbb{R}^{m \times n} \quad \begin{cases} m > n \text{ overdetermined} \\ m < n \text{ underdetermined} \end{cases}$$

$$\min_{\vec{x} \in \mathbb{R}^n} \overbrace{\|\vec{A}\vec{x} - \vec{b}\|_2^2}^{\text{residuals}} \implies \vec{x}^T A^T A \vec{x} - 2\vec{b}^T A \vec{x} - \vec{b}^T \vec{b}$$

$$\nabla_{\vec{x}}(\cdot) = 0 \implies A^T A \vec{x} = A^T \vec{b} \quad // \text{ Normal eqs}$$

$$\text{if } m > n, A^T A \succ 0 \implies \vec{x} = (A^T A)^{-1} A^T \vec{b}$$

$$f(x) = ax^2 + bx + c \implies \begin{bmatrix} x_1^2 & x_1 & 1 \\ \vdots & \vdots & \vdots \\ x_m^2 & x_m & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$$

Gradient Method

Converges linearly

$$\mathbf{def.} \quad \vec{d} \in \mathbb{R}^n \text{ is descent dir. of } f \text{ at } \vec{x} \text{ if } \begin{cases} \nabla f(\vec{x})^T \vec{d} < 0 \\ \nabla f(\vec{x}; \vec{d}) < 0 \end{cases}$$

$$\text{gradient w/ exact line search satisfies } f(\vec{x}_{k+1}) \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^2 f(\vec{x}_k), \quad \kappa = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$$

$$\text{if } f \in C^1(\mathbb{R}^n), \text{ bounded below, and } \nabla f(\vec{x}) \in C_L^{1,1}, L, M > 0 :$$

$$\text{as } k \rightarrow \infty \begin{cases} f \text{ flattens } \left\{ \begin{array}{l} f(\vec{x}_k) - f(\vec{x}_{k+1}) \geq M \|\nabla f(\vec{x}_k)\|^2 \geq 0 \\ \min_{k=0, \dots, n} \|\nabla f(\vec{x}_k)\| \leq \sqrt{\frac{f(\vec{x}_0) - f^*}{M(n+1)}} \end{array} \right. \end{cases}$$

Descent Direction "Method"

Inputs: $\vec{x}_0 \in \mathbb{R}^n$, tolerance $\varepsilon > 0$ While $\|\nabla f(\vec{x})\| > \varepsilon$ 1. Choose $\vec{d}_k$

$$\vec{d}_k = \begin{cases} -\nabla f(\vec{x}) & // \text{ gradient/steepest descent} \\ -[\nabla^2 f(\vec{x})]^{-1} \nabla f(\vec{x}) & // \text{ Newton (only if } H \succ 0) \end{cases}$$

2. Choose step size t_k via line search

$$t_k = \begin{cases} \tilde{t} & // \text{ constant} \\ \arg \min_{t \geq 0} f(\vec{x}_k + t_k \vec{d}_k) & // \text{ exact} \\ & // \text{ see backtrack} \end{cases}$$

3. Set $\vec{x}_{k+1} = \vec{x}_k + t_k \vec{d}_k, k = k + 1$

Backtracking Line Search

Inputs: $s > 0, \alpha \in (0, 1), \beta \in (0, 1)$ 1. Set $t_k = s$

$$2. \text{ While } f(\vec{x}_k + t_k \vec{d}_k) > f(\vec{x}_k) + \alpha t_k \nabla f(\vec{x}_k)^T \vec{d}_k$$

$$t_k = \beta t_k$$

Lipschitz Continuity

"If the next derivative is bounded by some L for all x, then the function is Lipschitz at the current derivative with L being the boundary".

$$\mathbf{def.} \quad f \in C_L^{1,1}(\mathbb{R}^n) \iff \|\nabla f(\vec{x}) - \nabla f(\vec{y})\| \leq L \|\vec{x} - \vec{y}\|$$

$$\text{thm 4.20} \quad f \in C_L^{1,1}(\mathbb{R}^n) \iff \|\nabla^2 f(\vec{x})\| \leq L, \forall \vec{x} \in \mathbb{R}^n$$

$$C^m < C_L^{a,b} < C^{m+1} \quad // \text{ Lipschitz is "in between"}$$

$$f \in C_L^{1,1}(\mathbb{R}^n) \implies f(\vec{y}) \leq f(\vec{x}) + \nabla f(\vec{x})^T (\vec{y} - \vec{x}) + \frac{L}{2} \|\vec{y} - \vec{x}\|^2$$

$$f \in C_L^{1,1}(\mathbb{R}^n) \implies f(\vec{x}) - f(\vec{x} - t \nabla f(\vec{x})) \geq t(1 - Lt/2) \|\nabla f(\vec{x})\|^2$$

(Pure) Newton's Method

Converges quadratically

Inputs: $\vec{x}_0 \in \mathbb{R}^n$, tolerance $\varepsilon > 0$ While $\|\nabla f(\vec{x})\| > \varepsilon$

1. Set $\vec{d}_k = -[\nabla^2 f(\vec{x}_k)]^{-1} \nabla f(\vec{x}_k)$
2. Set $\vec{x}_{k+1} = \vec{x}_k + \vec{d}_k$ // $t_k = 1$
3. Set $k = k + 1$

"Newton's method approximates a function as quadratic and solves it directly. If the function is quadratic already, it solves it in one step".

"Only descends if $A \succ 0$, and still can fail. Expensive, solves linear system (∇^2) at every k ".

Assume $\nabla^2 f(\vec{x}) \succ mI, \forall \vec{x} \in \mathbb{R}^n$, then:

$$\begin{cases} \|\nabla^2 f(\vec{x}) - \nabla^2 f(\vec{y})\| \leq L \|\vec{x} - \vec{y}\| & // \text{ Lipschitz} \\ \|\vec{x}_{k+1} - \vec{x}^*\| \leq \frac{L}{2m} \|\vec{x}_k - \vec{x}^*\|^2 & // \text{ Converge quad.} \end{cases}$$

Damped Newton's Method

Newton's Method but Backtracking for step size.

Def. Weierstrass Thm : if $f \in C^1(C \subseteq \mathbb{R}^n)$, C non-empty, compact : $\exists$ global (min, max) of f over C

2nd Half

min concave f(over convex set) $\implies \exists$ min at ext pt

In strictly convex opt prob, local min are global min
The set of opt pts to convex opt prob is also convex

Def. C convex $\iff \vec{x}, \vec{y}, [\lambda \vec{x} + (1-\lambda)\vec{y}] \in C, \lambda \in [0, 1]$

$$\underbrace{\text{if } S \subseteq \mathbb{R}^n, \vec{x} \in \text{conv}(S);}_{\text{Caratheodry thm}} \quad \begin{matrix} \exists \vec{x}_1, \dots, \vec{x}_{n+1} \in S \\ \text{s.t. } \vec{x} \in \text{conv}(\{\vec{x}_1, \dots, \vec{x}_{n+1}\}) \end{matrix}$$

Def. S is convex cone $\iff \begin{cases} \vec{x}, \vec{y} \in S \implies \vec{x} + \vec{y} \in S \\ \vec{x} \in S, \lambda \geq 0 \implies \lambda \vec{x} \in S \end{cases}$

Def. conic comb. $\lambda_1 \vec{x}_1 + \dots + \lambda_k \vec{x}_k$ w/ $\vec{x}_1, \dots, \vec{x}_n \in \mathbb{R}^n, \vec{\lambda} \in \mathbb{R}_+^k$

Def. $\vec{x} \in S$ is **ext pt** of $S \subseteq \mathbb{R}^n, \text{convex}$ if : $\nexists \vec{x}_1 \neq \vec{x}_2 \in S, \text{ s.t. } \vec{x} = \lambda \vec{x}_1 + (1-\lambda)\vec{x}_2$

if $S \subseteq \mathbb{R}^n$ is compact, convex : $S = \text{conv}(\text{ext}(S))$

Def. $f : C \rightarrow \mathbb{R}^n$ convex if :

$$f(\lambda \vec{x} + (1-\lambda)\vec{y}) \leq \lambda f(\vec{x}) + (1-\lambda)f(\vec{y}) \quad \forall \vec{x}, \vec{y} \in C, \lambda \in [0, 1]$$

Def. Jensen's ineq : $f(\sum_i \lambda_i \vec{x}_i) \leq \sum_i \lambda_i f(\vec{x}_i)$

$$\begin{cases} f \text{ convex} \iff f(\vec{x}) + \nabla f(\vec{x})^T (\vec{y} - \vec{x}) \leq f(\vec{y}) \quad \forall \vec{x}, \vec{y} \in C & 1^{\text{st}} \\ f \text{ convex} \iff \nabla^2 f(\vec{x}) \succeq 0 \quad \forall \vec{x} \in C & (\text{strictly if } \succ) & 2^{\text{nd}} \end{cases}$$

Def. Orthog. Proj: $P_c(\vec{x}) = \arg \min_{\vec{y} \in C} \|\vec{x} - \vec{y}\|^2$ // unique

Def. $\vec{x}^* \in C$ is **stat pt** if : $\nabla f(\vec{x}^*)^T (\vec{x} - \vec{x}^*) \geq 0 \quad \forall \vec{x} \in C$

$$\begin{cases} f \in C^1 \implies \text{local min are stat pts} & \text{nec for opt} \\ f \text{ also convex} & \text{suff for opt} \end{cases}$$

Gradient Projection Method Converges sublinearly
Set $\vec{x}_0 \in C, \varepsilon > 0$, and repeat for $k = 0, 1, \dots$

1. Pick step size t_k (small, or line search)

2. $\vec{x}_{k+1} = P_c(\vec{x}_k - t_k \nabla f(\vec{x}_k))$

3. If $\|\vec{x}_{k+1} - \vec{x}_k\| \leq \varepsilon$, STOP

Unlikely for $\nabla f(\vec{x}^*) \approx 0$. Different from other methods. Sub-linear converge if Lipschitz $\tilde{t}$ chosen well. (With $n = \#$ iters) KKT

$$f(\vec{x}_k) - f^* \leq \|\vec{x}_0 - \vec{x}^*\|^2 / (2\tilde{t}n)$$

$$\begin{aligned} (P) \quad & \min_{\vec{x} \in \mathbb{R}^n} f(\vec{x}) \quad f, g_i, h_j \in C^1(\mathbb{R}^n) \\ \text{s.t.} \quad & g_i(\vec{x}) \leq 0 \quad i = 1, \dots, m \\ & h_j(\vec{x}) = 0 \quad j = 1, \dots, p \end{aligned}$$

Def. $I(\vec{x}^*) = \{i \in [m] : g_i(\vec{x}^*) = 0\} =$ set of active constr.

Def. feasible pt $\vec{x}^*$ is **regular** if :

$\{\nabla g_i(\vec{x}^*) : i \in I(\vec{x}^*)\} \cup \{\nabla h_j(\vec{x}^*) : j = 1, \dots, p\}$ are lin. indep

Def. Slater's Condition holds if : $\exists \hat{\vec{x}} \in \mathbb{R}^n$ s.t. $g_i(\hat{\vec{x}}) < 0 \quad \forall i$

$$\begin{cases} \text{if } \vec{x}^* \text{ is local min of } (P), \text{ regular} & \implies \text{KKT pt} & \text{nec} \\ \quad (\text{Slater's cond holds}) & \\ \text{and if } (P) \text{ convex} & \implies \text{KKT pt IS opt pt} & \text{suf} \end{cases}$$

KKT Conditions

$$L(\vec{x}, \vec{\lambda}, \vec{\mu}) = f(\vec{x}) + \sum \lambda_i g_i(\vec{x}) + \sum \mu_j h_j(\vec{x})$$

$$\begin{aligned} \nabla_{\vec{x}} L &= \vec{0} & \lambda_i g_i(\vec{x}) &= 0 & \mu_j h_j(\vec{x}) &= 0 \\ g_i(\vec{x}) &\leq 0 & h_j(\vec{x}) &= 0 & \vec{\lambda} &\geq 0 \end{aligned}$$

Duality

$$q(\vec{\lambda}, \vec{\mu}) = \min_{\vec{x}} L(\vec{x}, \vec{\lambda}, \vec{\mu}) \quad // \text{always concave, sol is ext pt}$$

$$(D) \quad q^* = \max_{\vec{\lambda}, \vec{\mu}} q(\vec{\lambda}, \vec{\mu}) \quad // \text{always convex, even if } (P) \text{ isn't}$$

$$\text{dom}(q) = \{(\vec{\lambda}, \vec{\mu}) \in \mathbb{R}_+^m \times \mathbb{R}^p : q(\vec{\lambda}, \vec{\mu}) > -\infty\}$$

Weak Duality: $q^* \leq f^*$ always holds

Strong Duality: $q^* = f^*$ when (P) convex, w/ min $> -\infty$

Advanced Variants

"Dream: by approximating the Hessian instead of computing, the number of iterations can be reduced w/o incurring the cost of computing the Hessian".

- Diagonal scaling: Set $\vec{d}_k = D_k \nabla f(\vec{x}_k) = S S^T \nabla f(\vec{x}_k)$ where $\vec{x} = S \vec{y}$ is a change in variables to alter x_i 's, or non-dimensionalize to have similar sensitivity to changes across all x_i 's.
- BFGS Quasi-Newton: Alters the problem to utilize the inverse-Hessian while avoiding the issue of ever inverting. Can converge super-linearly (faster than grad. desc, slower than Newton's). Limited memory variant exist. Most opt solvers use this method.
- Gauss-Newton Method: Solves the non-linear least squares problem when $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ via Jacobian.

Content that didn't fit

$$\frac{(\vec{x}^T \vec{x})^2}{(\vec{x}^T A^{-1} \vec{x})} \geq \frac{4\kappa}{(\kappa+1)^2} \quad \kappa = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \quad // \text{Kantrovic ineq.}$$

Symmetric, Square A Properties

$$\text{Tr}(A) = \sum_{i=1}^n A_{ii} = \sum_{i=1}^n \lambda_i(A) \quad \det(A) = \prod_{i=1}^n \lambda_i(A)$$

$$R_A(\vec{x} \neq 0) = \frac{\vec{x}^T A \vec{x}}{\vec{x}^T \vec{x}} \quad \lambda_{\min}(A) \leq R_A(\vec{x} \neq 0) \leq \lambda_{\max}(A)$$

$$f \in C^2(\mathbb{R}^n) \quad (\nabla^2 f(\vec{x}))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j} \in \mathbb{R}^{n \times n}$$

$$h(\vec{x}) = f(g(\vec{x})) \quad \nabla h(\vec{x}) = \nabla g(\vec{x}) \nabla f(g(\vec{x})) \quad // \text{chain rule}$$

Sets

$$\text{int}(U) = \{\vec{x} \in U : B(\vec{x}, r) \subseteq U : \text{for some } r > 0\}$$

$$\text{Cl}(U) = \bigcap \{T : T \supseteq U, T \text{ closed}\}$$

$$U \subseteq \mathbb{R}^n \text{ bounded if } \exists M > 0 \text{ s.t. } U \subset B(\vec{0}, M)$$

$$\text{compact} = \text{closed} + \text{bounded}$$

$$S \subseteq \mathbb{R}^n \quad f^* = \inf\{f(\vec{x}) : \vec{x} \in S\} = \underbrace{\min\{f(\vec{x}) : \vec{x} \in S\}}_{\text{if proper min. in set}}$$

Definiteness of **Symmetric** Matrices

Name	Symbolic	Eigenvalues $\forall i$
PSD	$A \succeq 0$	$\lambda_i(A) \geq 0$
PD	$A \succ 0$	$\lambda_i(A) > 0$
NSD	$A \preceq 0$	$\lambda_i(A) \leq 0$
ND	$A \prec 0$	$\lambda_i(A) < 0$
Ind.	$A \not\preceq 0, A \not\succeq 0$	$\lambda_{\max}(A) > 0, \lambda_{\min}(A) < 0$

$$\text{Def. } A \in \mathbb{R}^{n \times n} \succeq 0 \quad \text{if } \vec{x}^T A \vec{x} \geq 0 \quad \forall \vec{x} \in \mathbb{R}^n$$

$$\text{Def. } A \in \mathbb{R}^{n \times n} \succ 0 \quad \text{if } \vec{x}^T A \vec{x} > 0 \quad \forall \vec{x} \in \mathbb{R}^n \setminus \{0\}$$

$$A \succ 0 \rightarrow A_{ii} > 0 \quad A \succeq 0 \rightarrow A_{ii} \geq 0$$

$$\text{if signs on diagonal differ, } A \text{ is Indef.}$$

Hybrid Gradient Newton Method

$$\text{if } \nabla^2 f(\vec{x}_k) \succ 0 \implies \vec{d}_k = -[\nabla^2 f(\vec{x}_k)]^{-1} \nabla f(\vec{x}_k)$$

$$\text{otherwise: } \vec{d}_k = -\nabla f(\vec{x}_k)$$

Circle Fitting

"Non-linear least squares problem that can be reformulated as a linear least squares problem".

$$\text{fit most pts. } \vec{a}_i \text{ with } C(\vec{x}, r) = \{\vec{a} \in \mathbb{R}^n : \|\vec{a} - \vec{x}\| = r\}$$

$$\text{Orig prob: } \min_{\vec{x}, R} \sum_{i=1}^m (-2\vec{a}_i^T \vec{x} + R + \|\vec{a}_i\|^2)^2$$

$$\text{Equivalent to: } \min_{\vec{y} \in \mathbb{R}^{n+1}} \|\tilde{A}\vec{y} - \vec{b}\|^2 \quad \text{where: } \vec{y} = \begin{bmatrix} \vec{x} \\ R \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$\tilde{A} = \begin{bmatrix} -2\vec{a}_1^T & -1 \\ \vdots & \\ -2\vec{a}_m^T & -1 \end{bmatrix} \in \mathbb{R}^{m \times n+1} \quad \vec{b} = \begin{bmatrix} \|\vec{a}_1\|^2 \\ \vdots \\ \|\vec{a}_m\|^2 \end{bmatrix} \in \mathbb{R}^m$$

$$\text{gives: } \vec{x}^* = y^*[1:n], \quad r^* = \sqrt{\|\vec{x}^*\|^2 - y^*[n+1]}$$